UNIT 2
LINEAR TRANSFORMATIONS

Overview

- Linear Transformation
- Null space and Range space
- Matrix representation of linear transformation
- Diagonalizations of linear transformation
- Application of diagonalization in linear system of differential equations.

LINEAR TRANSFORMATION

Let V and W be a vector space over a field F.

Then a function $\boldsymbol{T}: \boldsymbol{V} \rightarrow \boldsymbol{W}$ is said to be a linear transformation or linear mapping from V to W if for all

$x, y \in V \, and \, c \in F$

(a) $T(x+y) = T(x) + T(y)$

(b) $T(cx) = cT(x)$

Note:

- Identity transformation

$$I_V: V \to V \text{ by } I_V(x) = x \;\; \forall\, x \in V$$

- Zero transformation

$$T_0: V \to W \text{ by } T_0(x) = 0 \;\; \forall\, x \in V$$

- Derivative transformation

$$D: V \to V \text{ by } D(x) = \frac{dx}{dt} \;\; \forall\, x \in V, V = P(t) \; over \; R$$

- Integral transformation

$$J: V \to R \text{ by } J\big((f(t))\big) = \int_0^1 f(t)\, dt$$

PROPERTIES

1. If T is a linear, then $T(0) = 0$
2. T is linear $\Leftrightarrow$ $T(cx + y) = cT(x) + T(y)$
3. T is linear, then $T(x - y) = T(x) - T(y)$, for all $x, y \in V$
4. T is linear $\Leftrightarrow$ for $x_1, x_2 \dots . x_n \in V$ and $a_1, a_2 \dots a_n \in F$ we have

$$T\left(\sum_{i=1}^{n} a_i x_i\right) = \sum_{i=1}^{n} a_i \, T(x_i)$$

EXAMPLE 1 :

Let $T: R^2 \to R^2$ defined by $T(a_1, a_2) = (2a_1 + a_2, a_1)$. Show that T is linear Transformation.

SOLUTION

Given: $T: R^2 \to R^2$ defined by $T(a_1, a_2) = (2a_1 + a_2, a_1)$

Prove: T is linear Transformation.

Let $x = (a_1, a_2)$ and $y = (b_1, b_2)$ be in R and $c \in R$,

Then $x + y = (a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 + b_2)$

a) $T(x+y) \;=\; T\big((a_1+b_1, a_2+b_2)\big)$

$$= \big(2(a_1+b_1)+(a_2+b_2), (a_1+b_1)\big)$$

$$= \big((2a_1+2\,b_1+a_2+b_2), (a_1+b_1)\big) \qquad (1)$$

$$T(x) = T(a_1, a_2) = (2a_1+a_2, a_1)$$

$$T(y) = T(b_1, b_2) = (2b_1+b_2, b_1)$$

$$T(x) + T(y) = (2a_1+a_2, a_1) + (2b_1+b_2, b_1)$$

$$= (2a_1+a_2+2b_1+b_2, a_1+b_1) \qquad (2)$$

From (1) and (2) , $T(x+y) = T(x) + T(y)$

b) $T(cx) = T\big(c(a_1, a_2)\big)$

$= T(ca_1, ca_2)$

$= (2ca_1 + ca_2, ca_1) \quad (3)$

$c\,T(x) = c\,T(b_1, b_2)$

$= c\,(2b_1 + b_2, b_1)$

$= (2cb_1 + cb_2, cb_1) \quad (4)$

From (3) and (4) , $T(cx) = cT(x)$

Hence T is linear.

EXAMPLE 2 :

Let T: $P_n(R) \to P_{n-1}(R)$ defined by $T(f(x)) = f'(x)$. Prove that T is linear.

SOLUTION

Given that T: $P_n(R) \to P_{n-1}(R)$ defined by $T(f(x)) = f'(x)$

To prove: T is linear transformation.

Let $c \in R$ and let $f(x), g(x) \in P_n(R)$. Then

a) $T\big((f+g)(x)\big) = (f+g)'(x) = f'(x) + g'(x)$

$$= T\big(f(x)\big) + T\big(g(x)\big)$$

b) $T\big(cf(x)\big) = cf'(x) = cT\big(f(x)\big)$

$$\Rightarrow T\big(cf(x)\big) = cT\big(f(x)\big)$$

$\Rightarrow$ T is a linear transformation

EXAMPLE 3

Define $T : M_{3\times2}(F) \to M_{2\times3}(F)$ by $T(A) = A^t$ where A^t is the transpose of A . Then prove that T is a linear transformation.

SOLUTION

Given $T : M_{3\times2}(F) \to M_{2\times3}(F)$ by $T(A) = A^t$. Let A, $B \in M_{3\times2}(F)$ and $c \in F$, then

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} \qquad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}$$

$$A + B = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix} = \begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} \\ a_{21}+b_{21} & a_{22}+b_{22} \\ a_{31}+b_{31} & a_{32}+b_{32} \end{bmatrix}$$

$$(A + B)^t = \begin{bmatrix} a_{11}+b_{11} & a_{12}+b_{12} \\ a_{21}+b_{21} & a_{22}+b_{22} \\ a_{31}+b_{31} & a_{32}+b_{32} \end{bmatrix}^t$$

$$T(A+B) = \begin{bmatrix} a_{11}+b_{11} & a_{21}+b_{21} & a_{31}+b_{31} \\ a_{12}+b_{12} & a_{22}+b_{22} & a_{32}+b_{32} \end{bmatrix} \qquad (1)$$

$$T(A) = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}^t = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \end{bmatrix}$$

$$T(B) = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}^t = \begin{bmatrix} b_{11} & b_{21} & b_{31} \\ b_{12} & b_{22} & b_{32} \end{bmatrix}$$

$$\begin{aligned} T(A)+T(B) &= \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{21} & b_{31} \\ b_{12} & b_{22} & b_{32} \end{bmatrix} \\ &= \begin{bmatrix} a_{11}+b_{11} & a_{21}+b_{21} & a_{31}+b_{31} \\ a_{12}+b_{12} & a_{22}+b_{22} & a_{32}+b_{32} \end{bmatrix} \qquad (2) \end{aligned}$$

From (1) and (2), we have $T(A+B) = T(A) + T(B)$

$$\mathrm{T(A)} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}^t$$

$$\mathrm{T(cA)} = \begin{bmatrix} ca_{11} & ca_{12} \\ ca_{21} & ca_{22} \\ ca_{31} & ca_{32} \end{bmatrix}^t = \begin{bmatrix} ca_{11} & ca_{21} & ca_{31} \\ ca_{12} & ca_{22} & ca_{32} \end{bmatrix}$$

$$= c \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \end{bmatrix}$$

$$= cT(A).$$

Hence T is linear.

EXAMPLE 4 :

Define $T: \ R^2 \rightarrow R^2 \ by \ T(a_1, a_2) = (1, a_2)$ show that T is not a linear transformation.

SOLUTION

$$Let \ c \in R \ and \ x = (b_1, b_2) \ and \ y = (c_1, c_2) \ be \ in \ R, then$$

$$x + y = (b_1, b_2) + (c_1, c_2) = (b_1 + c_1, b_2 + c_2)$$

$$T(x + y) = T((b_1 + c_1, b_2 + c_2))$$

$$= (1, b_2 + c_2) \qquad (1)$$

$$T(x) = T(b_1, b_2) = (1, b_2)$$

$$T(y) = T(c_1, c_2) = (1, c_2)$$

$$T(x) + T(y) = (1, b_2) + (1, c_2)$$

$$= (2, b_2 + c_2) \qquad (2)$$

$$From \ (1) \ and \ (2), \quad T(x + y) \neq T(x) + T(y)$$

$$Hence \ T \ is \ not \ linear.$$

EXAMPLE 5 :

Define $T : R^2 \rightarrow R^2$ by $T(a_1, a_2) = (\sin\ a_1, 0)$ show that T is not a linear transformation.

SOLUTION

Let $c \in R$ and $x = (b_1, b_2)$ and $y = (c_1, c_2)$ be in R , then

$$x + y = (b_1, b_2) + (c_1, c_2) = (b_1 + c_1, b_2 + c_2)$$

$$T(x + y) = T\big((b_1 + c_1, b_2 + c_2)\big) = (\sin(b_1 + c_1), 0) \qquad (1)$$

$$T(x) = T(b_1, b_2) = (sin\ b_1, 0)$$

$$T(y) = T(c_1, c_2) = (\sin\ c_1, 0)$$

$$T(x) + T(y) = (\sin\ b_1, 0) + (\sin\ c_1, 0) = (\sin\ b_1 + \sin\ c_1, 0) \qquad (2)$$

From (1) and (2) , $T\,(x + y) \neq T(x) + T(y)$

Hence, it is not linear.

EXAMPLE 6 :

Suppose that $T:\ R^2 \to R^2$ is linear $T(1,0) = (1,4)$ and $T(1,1) = (2,5)$, what is $T(2,3) =$?

SOLUTION

Given $T:\ R^2 \to R^2$ is linear.

Now $(2,3) = a(1,0) + b(1,1) = (a+b, b)$

$a = -1 \quad , b = 3$

Then $T(2,3) = T(-1(1,0) + 3(1,1))$

$= -1T(1,0) + 3T(1,1)$

$= -1(1,4) + 3(2,5)$

$\mathrm{T}(2,3) = (5,11)$

Example 2

For any angle θ, define $\mathsf{T}_\theta \colon \mathsf{R}^2 \to \mathsf{R}^2$ by the rule: $\mathsf{T}_\theta(a_1, a_2)$ is the vector obtained by rotating (a_1, a_2) counterclockwise by θ if $(a_1, a_2) \neq (0, 0)$, and $\mathsf{T}_\theta(0, 0) = (0, 0)$. Then $\mathsf{T}_\theta \colon \mathsf{R}^2 \to \mathsf{R}^2$ is a linear transformation that is called the **rotation by** θ.

Example 3

Define $\mathsf{T} \colon \mathsf{R}^2 \to \mathsf{R}^2$ by $\mathsf{T}(a_1, a_2) = (a_1, -a_2)$. T is called the **reflection about the** x **-axis.** (See Figure 2.1(b).) ♦

Example 4

Define $\mathsf{T} \colon \mathsf{R}^2 \to \mathsf{R}^2$ by $\mathsf{T}(a_1, a_2) = (a_1, 0)$. T is called the **projection on the** x**-axis.** (See Figure 2.1(c).) ♦

Problems

1. Define $T: R^2 \to R^2$ by $T(a_1, a_2) = (a^2, a_2)$ show that T is not a linear transformation.

2. Define $T: R^2 \to R^2$ by $T(x, y) = (x + 2y, 2x - y, x + 5y)$ show that T is linear.

3. Define $T: R^3 \to R^2$ by $T(a_1, a_2, a_3) = (3a_1 - 2a_2 + a_3, -a_1 + 5a_2 + 4a_3)$ check whether that T is linear or not.

NULL SPACE, RANGE SPACE AND DIMENSION THEOREM

DEFINITION

Let V and W be vector spaces and let $T : V \to W$ be linear. We define the Null space (or kernel) $N(T) of\ T$ to be $the\ set\ of\ all\ vectors\ x\ in\ V\ such\ that\ T(x) = 0$

$$ie., N(T) = \{x \in V : T(x) = 0\}$$

DEFINITION

Let V and W be vector spaces and let $T : V \to W$ be linear. We define the Range Space (or image) *R(T)* of T be the subset of W consisting of all images (under T) of vectors in V

$$ie., R(T) = \{ T(x) : x \in V \}$$

DEFINITION

Let V and W be vector spaces and let $T : V \to W$ be linear. If $N(T)$ and $R(T)$ are finite dimensional, then

the nullity of T denoted by $\boldsymbol{nullity\ (T)}$ and is defined to be the dimension of $N(T)$ and

the rank of T is denoted by $\boldsymbol{rank(T)}$ and is defined to be dimension of $R(T)$ respectively.

i.e., $nullity(T) = dim\big(N(T)\big)$

$$rank(T) = dim(R(T))$$

NOTE

1. Let $I_v : V \rightarrow V$ be the identity on V.

   Then $N(I_v) = \{0\}$

   $R(I_v) = V$

   $nullity(I_v) = 0$

   $rank(I_v) = \dim(\text{V})$

2. Let $T_0 : V \rightarrow W$ zero transformation.

   Then $N(T_0) = V$

   $R(T_0) = \{0\}$

   $nullity(T_0) = 0$

   $rank(T_0) = \dim(\text{V})$

Example

Let $T: R^2 \to R^2$ defined by $T(a_1, a_2) = (a_1 + a_2, a_1)$. Then find $N(T)$

Solution

$$\begin{aligned} N(T) &= \{(a_1, a_2) \in \mathbb{R}^2 : T(a_1, a_2) = (0,0)\} \\ &= \{(a_1, a_2) \in \mathbb{R}^2 : (a_1 + a_2, a_1) = (0,0)\} \\ &= \{(a_1, a_2) \in \mathbb{R}^2 : a_1 = 0 = a_2\} \\ &= \{(0,0)\} \end{aligned}$$

THEOREM:

Let V and W be vector spaces and $T : V \to W$ be linear. Then $N(T)$ and $R(T)$ are subspaces V and W respectively.

PROOF

Given: V and W be vector spaces and $T : V \to W$ be linear.

To Prove: **$N(T)$ is a subspace of V.**

Let $O_V \in V$.

Then, $T(O_V) = O_W$

$\Longrightarrow \quad O_V \in N(T)$

Let $x, y \in N(T), and\ c \in F$. Then $T(x) = O_w, T(y) = O_w$

$$
\begin{aligned}
T(x + y) &= T(x) + T(y) \quad (Since\ T \text{ is linear}) \\
&= O_w + O_w \\
&= O_w
\end{aligned}
$$

$$\Rightarrow x + y \in N(T)$$

$$
\begin{aligned}
T(cx) &= cT(x) \\
&= cO_w \\
&= O_w
\end{aligned}
$$

$$\Rightarrow cx \in N(T)$$

$\therefore N(T)$ is a subspace $of\ V$

To Prove: $\boldsymbol{R(T)}$ is a subspace of $\boldsymbol{W}$

$T(O_v) = O_w \quad \Rightarrow \quad O_w \in R(T)$

Let $x, y \in R(T)$, $c \in F$. Then there exists v, w in V such that $T(v) = x \quad T(w) = y$

$T(v+w) = \quad T(v) + T(w)$ $(Since\ T$ is linear)

$\quad\quad = \quad x + y$

$\Rightarrow \quad x + y \in R(T)$

$T(cv) \quad = \quad cT(v)$

$\quad\quad = \quad cx$

$\Rightarrow \quad cx \in R(T)$

$\therefore\ R(T)$ is a subspace of W

THEOREM:

Let V and W be vector spaces and let $T: V \to W$ be linear. If $\beta = \{v_1, v_2, \ldots, v_n\}$ is a basis for V, then $R(T) = span\left(T(\beta)\right) = span\left\{\left(T(v_1), T(v_2), \ldots, T(v_n)\right)\right\}$

PROOF

Given $T: V \to W$ be linear

Then clearly $T(v_i) \in R(T)$ for each i.

We know that, $R(T)$ is subspace of W

By theorem "The span of any subset S of a vector space V is a subspace of V. Moreover, any subspace of V that contains S must also contain the $span\ of\ S$"

By the above fact, $R(T)$ contains $span(\{T(v_1), T(v_2), \ldots, T(v_n)\})$

$$\text{i.e., } span\,(\{T(v_1), T(v_2), \ldots, T(v_n)\}) \subseteq R(T) \qquad (1)$$

To prove that: $\boldsymbol{R(T) \subseteq \text{span}\ (T(\beta))}$

Now, suppose that $w \in R(T)$, then there exists $v \in V$ such that $T(v) = w$

Since $\beta = \{ v_1, v_2, \dots, v_n\}$ is a basis of V, then

$$
\begin{aligned}
v &= a_1 v_1 + a_2 v_2 + \cdots + a_n v_n \\
&= \textstyle\sum_{i=1}^{n} a_i v_i \qquad for\ some\ a_1, a_2, \dots \dots a_n \in F \\
\Rightarrow \quad T(v) &= T(\textstyle\sum_{i=1}^{n} a_i v_i) \\
\Rightarrow \quad w &= \textstyle\sum_{i=1}^{n} a_i T(v_i) \quad (\ since\ \ T\ is\ linear) \\
&= a_1 T(v_1) + a_2 T(v_2) + \cdots + a_n T(v_n) \\
&\in span\{T(v_1), T(v_2), \dots, T(v_n)\} \\
\therefore \quad w &\in span\left(T(\beta)\right)
\end{aligned}
$$

Hence $R(T) \subseteq span\left(T(\beta)\right)$ (2)

From (1) and (2) , we have $\boldsymbol{R(T)}$ **= span** $(\boldsymbol{T}(\beta)$

Note

If β is infinite basis, then $R(T)$ = span($\{T(v): v \in \beta\}$)

i.e., above theorem is true for infinite basis also

DIMENSION THEOREM

$$Let\ V\ and\ W\ be\ vector\ spaces\ and\ let\ T : V \to W\ be\ linear.\ If\ V\ is\ finite\ dimensional, then$$

$$\boldsymbol{nullity\ (T) + rank(T) = dim(V)}$$

PROOF

Let $dim(V) = n$ & $dim\left(N(T)\right) = k, consider\ \{v_1, v_2, \dots, v_k\}\ is\ a\ basis\ for\ N(T)$

By theorem , "If W is a subspace of finite dimensional vector space V, then any basis for W can be extended to a basis for V"

Using this fact, we may extend $\{v_1, v_2, \dots, v_k\}$ to a basis β = $\{v_1, v_2, \dots, v_k, v_{k+1}, \dots, v_n\}$ for V

$$\boldsymbol{i.e.,\ \beta = \{v_1, v_2, \dots, v_n\}\ \text{for}\ V}$$

Now, we claim that, $S = \{T(v_{k+1}), T(v_{k+2}), \dots, T(v_n)\}\ is\ a\ basis\ for\ R(T)$

i.e.,

i. S generates $R(T)$

ii. S is linearly independent

To prove: S generates $R(T)$.

By theorem, "*Let V and W* be vector spaces and let $T : V \to W$ *be linear. If* β = $\{v_1, v_2, \ldots, v_k\}$ is a basis for V, then $R(T) = span\left(T(\beta)\right) = span\left(\{T(v_1), T(v_2), \ldots, T(v_n)\}\right)$"

Using above theorem, we have $R(T) = span\left(\{T(v_1), T(v_2), \ldots\ldots T(v_n)\}\right)$ ------(*)

Since $\{v_1, v_2, \ldots, v_k\}$ be the basis for $N(T)$,

$\Rightarrow \quad T(v_i) = 0 \; for \; i = 1 \; to \; k. \quad i.e., T(v_1) = T(v_2) = \cdots = T(v_k) = 0$

From (*), we have $R(T) = Span(\{T(v_{k+1}), T(v_{k+2}), \ldots, T(v_n)\}$

i.e., $R(T) = span(S)$

Hence S generates $R(T)$

To prove: **S is linearly independent**

For the scalars $b_{k+1}, b_{k+2}, \dots, b_n \in F,$ we have

$$b_{k+1}T(v_{k+1}) + b_{k+2}\, T(v_{k+2}) + \cdots + b_n\, T(v_n) = 0 \qquad \text{----------} \quad (**)$$

$i.e.,$

$$\sum_{i=k+1}^{n} b_i T(v_i) = 0$$

$$\Longrightarrow \quad T\left(\sum_{i=k+1}^{n} b_i v_i\right) = 0 \qquad \text{[Since } T \text{ is linear]}$$

$$\sum_{i=k+1}^{n} b_i T(v_i) \in N(T)$$

Hence, there exists $c_1, c_2, \dots, c_k \in F$ *such that*

$$\sum_{i=k+1}^{n} b_i v_i = c_1 v_1 + c_2 v_2 + \cdots + c_k v_k$$

[since $\{v_1, v_2, \dots, v_k\}$ *is a basis for* $N(T)$]

$$\text{i.}e., \sum_{i=k+1}^{n} b_i v_i = \sum_{i=1}^{k} c_i v_i$$

$$\Rightarrow \sum_{i=k+1}^{n} b_i v_i - \sum_{i=1}^{k} c_i v_i = 0$$

$$\Rightarrow \sum_{i=1}^{k} (-c_i) v_i + \sum_{i=k+1}^{n} b_i v_i = 0$$

Since $\beta = \{v_1, v_2, \dots, v_n\}$ is a basis for V, we have $\{v_1, v_2, \dots, v_n\}$ are Linearly independent

$\Rightarrow \quad -c_i = 0$ for $i = 1$ to k

$b_i = 0$ for $i = k+1$ to n, $i.e.,\ b_{k+1} = b_{k+2} = \cdots = b_n = 0$

From *(**)*, we have $\{T(v_{k+1}), T(v_{k+2}), \dots \dots \dots T(v_n)\}$ is linear independent.

i.e., S is linear independent

Hence $S = \{T(v_{k+1}), T(v_{k+2}), \dots \dots \dots T(v_n)\}$ is a basis for $R(T)$

$\therefore dim\left(R(T)\right) = n - k,$

i.e., $\quad rank\ (T) = n - k$

$\Longrightarrow \quad rank\ (T) = \dim(V) - \dim\left(N(T)\right)$

$\Longrightarrow \quad rank\ (T) = dim(V) - nullity\ (T)$

$\Longrightarrow \quad \boldsymbol{rank\ (T) + nullity\ (T) = dim(V)}$

Hence the theorem.

THEOREM

Let V and W be vector spaces and let $T: V \to W$ be linear. Then T is $1-1 \Leftrightarrow N(T) = \{0\}$

PROOF

Given $T: V \to W$ linear.

Assume T is $1-1$, then we have to prove $N(T) = \{0\}$

Let $x \in N(T)$.

Then

$T(x) = 0 = T(0)$ [since T is linear]

$\Longrightarrow x = 0$ [since T is $1-1$]

Hence $N(T) = \{0\}$

Conversely,

Assume $N(T) = \{0\}$.

Then we have to prove T is $1 - 1$.

Suppose that $T(x) = T(y)$

$\Longrightarrow \quad T(x) - T(y) = 0$

$\Longrightarrow \quad T(x - y) = 0$ [since T is linear]

$\Longrightarrow \quad x - y \in N(T)$

$\Longrightarrow \quad x - y = 0$ [since $N(T) = \{0\}$]

$\Longrightarrow \quad x = y$

$\therefore \quad$ *T* is *1-1* Hence the theorem.

Example

Let $T : R^3 \to R^2$ be a linear transformation defined by $T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$. Then find $N(T), R(T), nullity(T), rank(T)$ and verify the dimension theorem. Also determine whether T is $1 - 1$ or $onto$.

SOLUTION

Given: $T : R^3 \to R^2$ be a linear defined by $T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$

Null space $N(T)$

Let $(a_1, a_2, a_3) \in N(T)$.

Then $T(a_1, a_2, a_3) = 0$

$\Longrightarrow \quad (a_1 - a_2 , \ 2a_3) = (0,0)$

$\Longrightarrow \quad a_1 - a_2 = 0, \ 2a_3 = 0$

$\Longrightarrow \quad a_1 = a_2, \ a_3 = 0$

$$\therefore \quad N(T) = \{(a, a, 0): a \in R\} = \{a(1,1,0): a \in R\}$$

Range space $R(T)$

We know that, $e_1 = (1,0,0), e_2 = (0,1,0), e_3 = (0,0,1)$ is a basis for R^3 and

$$R(T) = span\{T(e_1), T(e_2), T(e_3)\}$$

Now,

$T(e_1)= T(1,0,0)= (1-0,0) = (1,0)$

$T(e_2)= T(0,1,0)= (0-1,0) = (-1,0)$

$T(e_3)= T(0,0,1)= (0-0,2) = (0,2)$

$R(T) = span\{(1,0),(-1,0),(0,2)\}$

$\therefore$ $\{(1,0),(-1,0),(0,2)\}$ generates R^2.

To check Linear independent vectors

$$A = \begin{bmatrix} 1 & 0 \\ -1 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 2 \end{bmatrix} \; R_2 \to R_2 + R_1$$

$\therefore$ $\{(1,0),(0,2)\}$ is linearly independent.

Basis of $R(T) = \{(1,0),(0,2)\}$

Since, $N(T) = \{(a, a, 0): a \in R\} = \{a(1,1,0): a \in R\}$, *we have*

$nullity\ (T) = \dim\big(N(T)\big) = 1$

Since, $R(T) = \{(1,0), (0,2)\}$, *We have*

$rank(T) = \dim\big(R(T)\big) = 2$

Verification of dimension theorem

$rank\ (T) + nullity\ (T) = 1 + 2 = 3 = \dim(V)$

$\therefore$ Dimension theorem is verified.

Here, $N(T) \neq \{0\} \implies T\ is\ not\ 1 - 1$

Also, $dim\ (R(T)) = 2 = dim(R^2) \implies T$ is onto.

PROBLEMS

1. Prove that T is a linear transformation and find bases $N(T)$ & $R(T)$. Compute the *nullity & rank of* T and verify the dimension theorem, Also determine whether T is 1-1 or onto for the following. $T: P_2(R) \rightarrow P_3(R)$ *defined by* $T\big(f(x)\big) = xf(x) + f^{'}(x)$
2. If $T: R^4 \rightarrow R^3$ is a linear Transformation defined by

$T(x_1, x_2, x_3, x_4) = (x_1 - x_2 + x_3 + x_4, x_1 + 2x_3 - x_4,\ x_1 + x_2 + 3x_3 - 3x_4)\quad for\ the\ x_1, x_2, x_3, x_4 \in R$ then verify that dimension theorem.

PROBLEM

Let $T: P_2(R) \to P_3(R)$ be the linear transformation defined by $T(f(x)) = 2\,f^{'}(x) + 3\,\int_0^x f(t)dt$.

Check whether T is $1 - 1$ or not.

SOLUTION

Since $\beta = \{1, x, x^2\}$ is a basis for $P_2(R)$

We know that $R(T) = span\left(T(\beta)\right)$

$$i.e., \; R(T) = span\,(\{T(1), T(x), T(x^2)\})$$

Now

$T(1) = 2(0) + 3\,\int_0^x dt = 3(t)_0^x = 3x$

$T(x) = 2(1) + 3\,\int_0^x tdt = 2 + 3\left(\frac{t^2}{2}\right)_0^x = 2 + \frac{3x^2}{2}$

$T(x^2) = 2(2x) + 3\,\int_0^x t^2dt = 4x + 3\left(\frac{t^3}{3}\right)_0^x = 4x + \mathrm{x}^3$

To check $\{3x, 2 + \frac{3x^2}{2}, 4x + x^3\}$ is linear independent.

For scalars, $\alpha, \beta, \gamma \in F$ *we have*

$$\alpha(3x) + \beta\left(2 + \frac{3x^2}{2}\right) + \gamma\,(4x + x^3) = 0$$

$$\Longrightarrow \qquad \gamma\, x^3 + \frac{3}{2}\beta\, x^2 + (3\alpha + 4\gamma)x + 2\beta = 0$$

$$\Longrightarrow \qquad \gamma = 0\,,\ \frac{3}{2}\beta = 0, (3\alpha + 4\gamma) = 0,\ 2\beta = 0$$

$$\Longrightarrow \qquad \alpha = 0, \beta = 0, \gamma = 0$$

$\therefore\ \{3x, 2 + \frac{3x^2}{2}, 4x + x^3\}$ is linear independent

Hence $\{3x, 2 + \frac{3x^2}{2}, 4x + \mathrm{x}^3\}$ is a basis for $R(T)$.

$\dim(R(T)) = rank\ (T) = 3$ & $\dim(P_3(R)) = 4$

$\therefore rank\ (T) \neq \dim(P_3(R))$

$\Longrightarrow T$ is not $onto$.

By dimension theorem, we have

$rank(T) + nullity(T) = \dim(V)$,

$$\Longrightarrow \quad 3 + nullity(T) \quad = \quad 3$$

$$\Longrightarrow \quad nullity(T) \quad = \quad 0$$

$$\Longrightarrow \quad N(T) \quad = \quad \{0\}$$

By theorem " Let V and W be vector spaces and let $T : V \to W$ $be\ linear$. Then T is $1-1 \Leftrightarrow N(T) = \{0\}$"

$\therefore$ T is $1-1$.

THEOREM

Let V and W be vector space of equal (finite) dimension and let $T : V \to W$ be linear. Then the following are equivalent :

a. T is $1 - 1$

b. $T\ is\ onto$

c. $rank\ (T) = \dim(V)$

PROOF

From the dimension theorem, we have

$nullity\ (T) + rank\ (T) = \dim(V)$ ---------- *(1)*

By the theorem, "Let V and W be vector spaces and let $T: V \to W$ be linear. Then T is $1 - 1 \Leftrightarrow N(T) = \{0\}$"

Using the above fact

$$
\begin{aligned}
T \text{ is } 1-1 \quad &\Leftrightarrow \quad N(T) = \{0\} \\
&\Leftrightarrow \quad nullity\ (T) = 0 \\
&\Leftrightarrow \quad rank\ (T) = dim(V) \qquad [\therefore \text{using } (1)] \\
&\Leftrightarrow \quad dim(R(T)) = dim(W)
\end{aligned}
$$

By the theorem, "Let V and W be vector spaces of finite dimensional vctor space V, then W is finite dimensional and $\dim(W) \leq \dim(V)$. Moreover, if $\dim(W) = \dim(V)$, then $V = W$"

By the above fact, $R(T) = W$

$$
\begin{aligned}
T \text{ is } 1-1 \quad &\Leftrightarrow \quad dim(R(T)) = dim(W) \\
&\Leftrightarrow \quad R(T) = W \\
&\Leftrightarrow \quad T \text{ is onto.}
\end{aligned}
$$

PROBLEM

Let $T: P_2(R) \to P_3(R)$ be the linear transformation defined by $T(f(x)) = 2\,f'(x) + 3\,\int_0^x f(t)dt$.

Check whether T is $1-1$ or not.

SOLUTION

Since $\beta = \{1, x, x^2\}$ is a basis for $P_2(R)$

We know that $R(T) = span\left(T(\beta)\right)$

$$i.e.,\ R(T) = span\,(\{T(1), T(x), T(x^2)\})$$

Now

$T(1) = 2(0) + 3\int_0^x dt = 3(t)_0^x = 3x$

$T(x) = 2(1) + 3\int_0^x t\,dt = 2 + 3\left(\frac{t^2}{2}\right)_0^x = 2 + \frac{3x^2}{2}$

$T(x^2) = 2(2x) + 3\int_0^x t^2 dt = 4x + 3\left(\frac{t^3}{3}\right)_0^x = 4x + x^3$

To check $\{ 3x, 2 + \frac{3x^2}{2}, 4x + x^3\}$ is linear independent.

For scalars, $\alpha, \beta, \gamma \in F$ *we have*

$$\alpha(3x) + \beta \left(2 + \frac{3x^2}{2}\right) + \gamma\,(4x + x^3) = 0$$

$$\Longrightarrow \qquad \gamma\, x^3 + \frac{3}{2}\beta\, x^2 + (3\alpha + 4\gamma)x + 2\beta = 0$$

$$\Longrightarrow \qquad \gamma = 0\,,\ \frac{3}{2}\beta = 0, (3\alpha + 4\gamma) = 0,\ 2\beta = 0$$

$$\Longrightarrow \qquad \alpha = 0, \beta = 0, \gamma = 0$$

$\therefore\ \{ 3x, 2 + \frac{3x^2}{2}, 4x + x^3\}$ is linear independent

Hence $\{ 3x, 2 + \frac{3x^2}{2}, 4x + \mathrm{x}^3\}$ is a basis for $R(T)$.

$\dim\big(R(T)\big) = rank\ (T) = 3\ \ \&\ \ \dim\big(P_3(R)\big) = 4$

$\therefore rank\ (T) \neq \dim\big(P_3(R)\big)$

$\Longrightarrow T$ is not $onto$.

By dimension theorem, we have

$rank(T) + nullity(T) = \dim(V)$,

$$\Longrightarrow \quad 3 + nullity(T) = 3$$

$$\Longrightarrow \quad nullity(T) = 0$$

$$\Longrightarrow \quad N(T) = \{0\}$$

By theorem " Let V and W be vector spaces and let $T : V \to W$ *be linear*. Then T is $1-1 \Leftrightarrow N(T) = \{0\}$"

$\therefore\ T$ is $1-1$.

PROBLEM

Let $T: F^2 \rightarrow\ F^2$ be linear transformation defined by $T(a_1, a_2) =\ (a_1 + a_2, a_1)$, then prove that T is onto.

SOLUTION

Given $T: F^2 \rightarrow\ F^2$ is linear.

Let $(a_1, a_2) \in F^2$. Then

$$T(a_1, a_2) =\ (a_1 + a_2, a_1)$$

$$\begin{aligned} N(T) &= \{\ (a_1, a_2): (a_1 + a_2, a_1) = 0\} \\ &= \{\ (a_1, a_2):\ a_1 = 0, a_2 = 0\} \\ &=\ \{0\} \end{aligned}$$

$\therefore nullity\ (T)\ =\ 0$

By theorem " Let V and W be vector spaces and let $T : V \rightarrow W$ *be linear*. Then T is $1 - 1 \iff N(T) =\ \{0\}$"

$\therefore\ T$ is $1 - 1$.

By theorem " Let V and W be vector space of equal (finite) dimension and let $T : V \rightarrow W$ be linear. Then the following are equivalent : a. T is $1 - 1$ b. T *is onto* c. $rank\ (T)\ =\ \dim(V)$ "

$\therefore\ T$ is *onto*

PROBLEM 1

Let $T: P_2(R) \to R^3$ defined by $T(a_0 + a_1x + a_2x^2) = (a_0, a_1, a_2)$, then prove that T is linear transformation and one-one.

PROBLEM 2

Let $T: \mathrm{R}^3 \to R^2$ be the linear defined by $T(a_1, a_2, a_3) = (a_1 - a_2, 2a_3)$, then prove that $T\ is\ onto$.

THEOREM

Let V and W be vector spaces over F and suppose that $\{v_1, v_2, \ldots, v_n\}$ is a basis for V . For $w_1, w_2, \ldots, w_n \in W$

there exist exactly one linear transformation $T : V \rightarrow W$ such that $T(v_i)= w_i$ for $i = 1,2, \ldots, n$.

PROOF

Let $x \in V$, then $x = \sum_{i=1}^{n} a_i v_i$, where $a_1, a_2, \ldots, a_n$ are unique scalars.

Define $T: V \rightarrow W$ by $T(x) = \sum_{i=1}^{n} a_i w_i$

To prove : *T* is linear

Suppose that $x, y \in V$ & $c \in F$.

Then $x = \sum_{i=1}^{n} a_i v_i$ & $y = \sum_{i=1}^{n} b_i v_i$ for some scalars an $a_1, a_2, \ldots, a_n , b_1, b_2 , \ldots, b_n$

$$
\begin{aligned}
\mathrm{c}x + y &= c\sum_{i=1}^{n} a_i v_i + \sum_{i=1}^{n} b_i v_i \\
&= \sum_{i=1}^{n} c a_i v_i + \sum_{i=1}^{n} b_i v_i \\
&= \sum_{i=1}^{n} (c a_i + b_i) v_i
\end{aligned}
$$

$$
\begin{aligned}
\Longrightarrow \quad T(cx + y) &= T\left(\sum_{i=1}^{n} (c a_i + b_i) v_i\right) \\
&= \sum_{i=1}^{n} (c a_i + b_i) w_i \\
&= \sum_{i=1}^{n} c a_i w_i + \sum_{i=1}^{n} b_i w_i \\
&= c\sum_{i=1}^{n} a_i w_i + \sum_{i=1}^{n} b_i w_i \\
&= cT(x) + T(y)
\end{aligned}
$$

$\therefore$ T is linear Transformation.

To prove T is unique

Suppose that $U: V \rightarrow W$ is a another linear trasformation and $U(v_i) = w_i$ for $i = 1, 2, \dots, n$

Then for $x \in \mathrm{V}$ with $x = \sum_{\mathrm{i}=1}^{\mathrm{n}} \mathrm{a_i v_i}$

We have

$$U(x) = \sum_{i=1}^{n} a_i U(v_i)$$

$$= \sum_{i=1}^{n} a_i w_i$$

$$= T(x)$$

Hence $U = T$

$\therefore$ T is unique linear transformation.

Corollary:

Let V and W be vector spaces and suppose that V has a finite basis $\{\mathrm{v}_1, \mathrm{v}_2, \dots, \mathrm{v}_\mathrm{n}\}$.

If $U, T: V \to W$ are linear & $U(v_i) = T(v_i)$ for $i = 1,2, \dots, n$, then $U = T$.

Example

Let T: $\mathrm{R}^2 \to R^2$ be the linear defined by $T(a_1, a_2) = (2a_2 - a_1, 3a_1)$.

Suppose that $U: R^2 \to R^2$ is linear.

If, $U(1,2) = (3,3)$ and $U(1,1) = (1,3)$, then $U = T$

From the above corollary, $\{(1,2), (1,1)\}$ is a basis for R^2.

MATRIX REPRESENTATION OF LINEAR TRANSFORMATIONS

DEFINITION

Let V be a finite-dimensional vector space. An ordered basis for V is a basis for V with a specific order.

i.e., an ordered basis for V is a finite sequence of linearly independent vectors in V that generatres V.

Example

1) $\beta = \{e_1, e_2, \ldots, e_n\}$ is the standard ordered basis for vector space F^n.

2) $\beta = \{1, x, \ldots, x^n\}$ is the standard ordered basis for vector space $P_n(F)$.

3) In R^3, $\beta = \{e_1, e_2, e_3\}$ an ordered basis. Also, $\gamma = \{e_2, e_1, e_3\}$ is an ordered basis but

   $\beta \neq \gamma$ as ordered bases.

DEFINITION

Let $\beta = \{u_1, u_2, \dots, u_n\}$ be an ordered basis for a finite - dimensional vector space V.

For $x \in V$, let $a_1, a_2, \dots, a_n$ be the unique scalars such that $x = \sum_{i=1}^{n} a_i u_i$.

Then the co-ordinate vector of x relative to β, denoted $[x]_\beta$ and defined by

$$[x]_\beta = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$$

Example

Let $V = P_2(R)$and let $\beta = \{1, x, x^2\}$ be the standard ordered basis for V,

If $f(x) = 4 + 6x - 7x^2$, then $[f]_\beta = \begin{bmatrix} 4 \\ 6 \\ -7 \end{bmatrix}$

MATRIX REPRESENTATION OF A LINEAR TRANSFORMATION

Let V and W are finite-dimensional vector space over F, with ordered bases $\beta = \{v_1, v_2, \dots, v_n\}$ and

$\gamma = \{w_1, w_2, \dots . w_n\}$ respectively. Let $T: V \to W$ be linear transformation.

Then for each $j, 1 \leq j \leq n$, there exist unique scalars $a_{ij} \in F, 1 \leq i \leq m$ such that

$$T(v_j) = \sum_{i=1}^{n} a_{ij} w_i \ \ for \ \ 1 \leq j \leq n$$

i.e.,

$$T(v_1) = \sum_{i=1}^{m} a_{i1} w_i = a_{11}w_1 + a_{21}w_2 + \cdots + a_{m1}w_m$$

$$T(v_2) = \sum_{i=1}^{m} a_{i2} w_i = a_{12}w_1 + a_{22}w_2 + \cdots + a_{m2}w_m$$

$$\vdots$$

$$T(v_n) = \sum_{i=1}^{m} a_{in} w_i = a_{1n}w_1 + a_{2n}w_2 + \cdots + a_{mn}w_m$$

The $m \times n$ matrix A defined by $A_{ij} = a_{ij}$, is called the matrix representation of T in the ordered bases β

and γ and denoted by $A = [T]_\beta^\gamma$.

i.e., $[T]_\beta^\gamma = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$

$\boldsymbol{Note}$: $If\ \beta = \gamma$, then $[T]_\beta{}^\gamma = [T]_\beta$

PROBLEM 1:

Let β and γ be the standard bases for R^2 and R^3 respectively. Then for each linear transformation $T: R^2 \rightarrow R^3$ defined by $T(a_1, a_2) = (2a_1 - a_2, 3a_1 + 4a_2, a_1)$, then compute $[T]_\beta^\gamma$.

SOLUTION:

Here $\beta = \{(1,0), (0,1)\}$ *and* $\gamma = \{(1,0,0), (0,1,0), (0,0,1)\}$

Given $T(a_1, a_2) = (2a_1 - a_2, 3a_1 + 4a_2, a_1)$

$$
\begin{aligned}
\text{Then} \quad T(e_1) &= T(1,0) \\
&= (2-0, 3+0, 1) \\
&= (2,3,1) \\
&= 2e_1' + 3e_2' + e_3' \\
T(e_2) &= T(0,1) \\
&= (0-1, 0+4, 0) \\
&= (-1,4,0) \\
&= -1e_1' + 4e_2'
\end{aligned}
$$

$$[T]_\beta^\gamma = \begin{bmatrix} 2 & -1 \\ 3 & 4 \\ 1 & 0 \end{bmatrix}$$

PROBLEM 2:

Let β and α be the standard bases for$P_2(R)$ and $M_{2\times 2}(R)$respectively. Then for each linear transformation $T: P_2(R) \to M_{2\times 2}(R)$ defined by $T\big(f(x)\big) = \begin{bmatrix} f'(0) & 2f(1) \\ 0 & f''(3) \end{bmatrix}$, then compute $[T]_\beta^\alpha$

SOLUTION:

Given $\beta = \{1, x, x^2\}$ and $\alpha = \left\{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}\right\}$

Then $T(1) = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = 0\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + 2\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + 0\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + 0\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

$T(x) = \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} = 1\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + 2\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + 0\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + 0\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

$T(x^2) = \begin{pmatrix} 0 & 2 \\ 0 & 2 \end{pmatrix} = 0\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + 2\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + 0\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} + 2\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$

Hence the required matrix is

$$[T]_\beta^\alpha = \begin{bmatrix} 0 & 1 & 0 \\ 2 & 2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

PROBLEM 3:

Find the linear transformation $T: V_3(R) \to V_3(R)$ determined by the matrix $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$ with respect to standard basis $\{e_1, e_2, e_3\}$.

SOLUTION:

Given $T: V_3(R) \to V_3(R)$ is linear and $[T]_\beta^\gamma = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ -1 & 3 & 4 \end{bmatrix}$

$\therefore$ coefficient matrix is $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ -1 & 3 & 4 \end{bmatrix}^t$

The standard basis are $\{e_1 = (1,0,0), e_2 = (0,1,0), e_3 = (0,0,1)\}$

$$T[e_1] = (1,0,-1) = e_1 + 0e_2 - e_3$$
$$T[e_2] = (2,1,3) = 2e_1 + e_2 + 3e_3$$
$$T[e_3] = (1,1,4) = e_1 + e_2 + 4e_3$$

Let $(a,b,c) \in V_3(R)$ be any element , then

$$
\begin{aligned}
(a,b,c) &= ae_1 + be_2 + ce_3 \\
T(a,b,c) &= aT(e_1) + bT(e_2) + cT(e_3) \\
&= a(1,0,-1) + b(2,1,3) + c(1,1,4) \\
&= (a + 2b + c, b + c, -a + 3b + 4c)
\end{aligned}
$$

which is the required transformations.

PROBLEM 4

Let $T: R^2 \to R^3$ be defined by $\mathrm{T}(\mathrm{a}_1, \mathrm{a}_2) = (\mathrm{a}_1 - \mathrm{a}_2, \mathrm{a}_1, 2\mathrm{a}_1 + \mathrm{a}_2)$. Let β be the standard ordered basis for R^2, given that $\gamma = \{(1,1,0), (0,1,1), (2,2,3)\}$ and $\alpha = \{(1,2),(2,3)\}$ then compute $[T]_\alpha{}^\gamma$.

Solution

Given $\alpha = \{(1,2),(2,3)\}$ and $\gamma = \{(1,1,0), (0,1,1), (2,2,3)\}$

$\mathrm{T}(1,2) = (-1,1,4)$

$(-1,1,4) = \alpha(1,1,0) + \beta(0,1,1) + \gamma(2,2,3)$

$\alpha + 2\gamma = -1$

$\alpha + \beta + 2\gamma = 1$

$\beta + 3\gamma = 4$

Solving the above equation , we get $\beta = 2$, $\gamma = \frac{2}{3}$ and $\alpha = -\frac{7}{3}$

$\mathrm{T}(2,3) = (-1,2,7)$

$(-1,2,7) = \alpha(1,1,0) + \beta(0,1,1) + \gamma(2,2,3)$

$\alpha + 2\gamma = -1$

$\alpha + \beta + 2\gamma = 2$

$\beta + 3\gamma = 7$

Solving the above equation , we get $\beta = 3$, $\gamma = \frac{4}{3}$ and $\alpha = -\frac{11}{3}$

$$\mathrm{T}(1,2) = -\frac{7}{3}(1,1,0) + 2(0,1,1) + \frac{2}{3}(2,2,3)$$

$$\mathrm{T}(2,3) = -\frac{11}{3}(1,1,0) + 3(0,1,1) + \frac{4}{3}(2,2,3)$$

$$[T]_{\alpha}{}^{\gamma} = \begin{bmatrix} -\frac{7}{3} & -\frac{11}{3} \\ 2 & 3 \\ \frac{2}{3} & \frac{4}{3} \end{bmatrix}$$

Example

Let $T: R^2 \to R^3$ be defined by $T(\mathrm{a}_1, \mathrm{a}_2) = (\mathrm{a}_1 - \mathrm{a}_2, \mathrm{a}_1, 2\mathrm{a}_1 + \mathrm{a}_2)$.

Let β be the standard ordered basis for $R^2, \gamma = \{(1,1,0), (0,1,1), (2,2,3)\}$ and

$\alpha = \{(1,2),(2,3)\}$, then compute $[T]_\beta^\gamma$ and $[T]_\alpha^\gamma$.

Solution

Let $\beta = \{e_1, e_2\}$, where $e_1 = (1,0)$, $e_2 = (0,1)$ and

$\gamma = \{e_1', e_2', e_3'\}$, where $e_1' = (1,1,0), e_2' = (0,1,1), e_3' = (2,2,3)$

Then, $T(1,0) = (1,1,2)$

$(1,1,2) = a(1,1,0) + b(0,1,1) + c(2,2,3)$

$\implies$ $a + 2c = 1$

$a + b + 2c = 1$

$b + 3c = 2$

Solving the above equation, we get $a = -\frac{1}{3}$, $b = 0$ and $c = \frac{2}{3}$

$\therefore$ $T(1,0) = -\frac{1}{3}e_1' + \frac{2}{3}e_3'$

$T(0,1) = (-1,0,1)$

$(-1,0,1) = a'(1,1,0) + b'(0,1,1) + c'(2,2,3)$

$$\Longrightarrow \qquad \begin{aligned} a' + 2c' &= -1 \\ a' + b' + 2c' &= 0 \\ b' + 3c' &= 1 \end{aligned}$$

Solving the above equation, we get $a' = -1$, $b' = 1$ and $c' = 0$

$\therefore \qquad T(0,1) = -e_1' + e_2'$

Hence, $\quad [T]_\beta^\gamma = \begin{bmatrix} -\frac{1}{3} & -1 \\ 0 & 1 \\ \frac{2}{3} & 0 \end{bmatrix}$

Let $\gamma = \{e_1', e_2', e_3'\}$, where $e_1' = (1,1,0), e_2' = (0,1,1), e_3' = (2,2,3)$

$\alpha = \{e_1'',\ e_2''\}$, where $e_1'' = (1,2),\ e_2'' = (2,3)$

Then, $T(1,2) = (-1,1,4)$

$(-1,1,4) = A(1,1,0) + B(0,1,1) + C(2,2,3)$

$\Longrightarrow \quad \mathrm{A} + 2C = -1$

$\mathrm{A} + B + 2C = 1$

$B + 3C = 4$

Solving the above equation , we get $\mathrm{A} = -\frac{7}{3},\ B = 2$ and $C = \frac{2}{3}$

$\therefore \quad T(1,2) = -\frac{7}{3}e_1' + 2e_2' + \frac{2}{3}e_3'$

$T(2,3\ ) = (-1,2,7\ )$

$(-1,2,7\ ) = A'(1,1,0) + B'(0,1,1) + C'(2,2,3)$

$$\Longrightarrow \qquad A' + 2B' = -1$$

$$A' + B' + 2C' = 2$$

$$B' + 3C' = 7$$

Solving the above equation , we get $A' = -\frac{11}{3}, B' = 3$ and $C' = \frac{4}{3}$

$$\therefore \qquad T(2,3\ ) = -\frac{11}{3}e_1' + 3e_2' + \frac{4}{3}e_3'$$

Hence, $[T]_\alpha^\gamma == \begin{bmatrix} -\frac{7}{3} & -\frac{11}{3} \\ 2 & 3 \\ \frac{2}{3} & \frac{4}{3} \end{bmatrix}$

DIAGONALIZATION OF LINEAR TRANSFORMATIONS

Definition

If V is a vector spaces over a field F, then a linear transformation T from V into V is called **linear operator on V**.

i.e., $T : V \to V$ is linear operator

Definition

If V is a vector spaces over a field F, then a linear transformation f from V into the scalar field F is called **linear functional on V**.

i.e., $f : V \to F$ is linear operator

Definition

Let V and W be vector spaces and let $T : V \rightarrow W$ be linear.

A function $U: W \rightarrow V$ is said to be an **inverse of *T*** if $TU = I_W$ and $UT = I_V$.

If T has an inverse, then T is said to be **invertible.**

If T is invertible, then the inverse of T is unique and is denoted by T^{-1}.

Note

T is invertible $\Leftrightarrow$ $T\ is\ 1 - 1$ and T is onto.

Definition

Let A be an $m \times n$ matrix with entries from a field F. Then the mapping

$L_A : F^n \to F^m$ defined by $L_A(x) = Ax$ for each column vector $x \in F^n$ is called the **left-multiplication transformation** and is denoted by L_A.

Example

Let $L_A: R^3 \to R^2$ and $A = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix} \in M_{2\times3}(R)$.

If $x = \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix}$, then $L_A = Ax = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 6 \\ 1 \end{bmatrix}$

Definition

A linear operator T on a finite $-$ dimensional vector space V is called **diagonalizable** if there is an ordered basis β for V such that $[T]_\beta$ is a diagonal matrix.

Note

A square matrix A is called diagonalizable if L_A is diagonalizable.

Note

If $D = [T]_\beta$ is a diagonal matrix, then for each vector $v_j \in \beta$, we have

$$T(v_j) = \sum_{i=1}^{n} D_{ij} v_i = D_{jj} v_j = \lambda_j v_j$$

where, $\lambda_j = D_{jj}$.

Conversely,

If $\beta = \{v_1, v_2, \dots, v_n\}$ is an ordered basis for V such that $T(v_j) = \lambda_j v_j$, for some scalars $\lambda_1, \lambda_2, \dots, \lambda_n$, then

$$[T]_\beta = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$$

Definition

Let T be an operator on a vector space V. A non-zero vector $v \in V$ is called an **eigenvector** of T if there exist a scalar λ such that $T(v) = \lambda v$.

The scalar λ is called **eigenvalue** corresponding to the eigen vector v.

Definition

Let $A \in M_{m \times n}(F)$. A non-zero vector $v \in F^n$ is called an eigenvector of A if v is an eigenvector of L_A. i.e., $Av = \lambda v$ for some scalar λ.

The scalar λ is called the eigenvalue of A corresponding to the eigenvector v.

Definition

Let T be a linear operator on a vector space V and let λ be an eigenvalue of T. Then, the set E_λ is called the **Eigen space** of T corresponding to the eigenvalue λ is define by

$$\boldsymbol{E_\lambda = \{x \in V: T(x) = \lambda x\} = N(T - \lambda I_V).}$$

We define the Eigen space of a square matrix A to be the Eigen space of L_A.

Note

- Eigen vector is also called *characteristic vector* or proper vector.
- Similarly Eigen value is also called as characteristic value or proper value.

Theorem

A linear operator T on a finite-dimensional vector space V is diagonalizable if and only if there exists an ordered basis β for V consisting of eigenvectors of T.

Furthermore, if T is diagonalizable, $\beta = \{v_1, v_2, \dots, v_n\}$ is an ordered basis of eigenvectors of T and $D = [T]_\beta$, then D is a diagonal matrix and D_{jj} is the eigenvalue corresponding to v_j for $1 \leq j \leq n$.

Note

To diagonalizable a matrix or linear operator, we have to find a basis of eigen vectors and the corresponding eigen values.

Example

Let $A = \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix}$, $\beta = \{v_1, v_2\}$, where $v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$. Then check A is diagonalizable or not?

Solution

$$
\begin{aligned}
L_A(v_1) &= \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\
&= \begin{bmatrix} -2 \\ 2 \end{bmatrix} \\
&= -2 \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\
&= -2v_1
\end{aligned}
$$

$\therefore$ v_1 is a eigenvector of L_A

Hence, v_1 is a eigenvector of A and the corresponding eigenvalue is $\lambda_1 = -2$

$$L_A(v_2) = \begin{bmatrix} 1 & 3 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$= \begin{bmatrix} 15 \\ 20 \end{bmatrix}$$

$$= 5 \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$= 5v_2$$

$\therefore$ v_2 is a eigenvector of L_A

Hence, v_2 is a eigenvector of A and the corresponding eigenvalue is $\lambda_2 = 5$

HOME WORK: $\beta = \{v_1, v_2\}$ is an ordered basis for R^2.

Also, β consisting of eigenvectors of both A and L_A.

By theorem "A linear operator T on a finite-dimensional vector space V is diagonalizable if and only if there exists an ordered basis β for V consisting of eigenvectors of T"

$\therefore$ A and L_A are diagonalizable.

$$D = \begin{bmatrix} -2 & 0 \\ 0 & 5 \end{bmatrix}$$

Definition

Let $A \in M_{n\times n}(F)$, then the polynomial $f(t) = det(A - tI_n)$ is called the **characteristic polynomial** of A.

Definition

Let T be a linear operator on an n-dimensional vector space V with ordered basis β. We define the **characteristic polynomial of T** to be the characteristic polynomial of $A = [T]_\beta$.

i.e., $f(t) = \det(A - tI_n)$

Theorem

Let $A \in M_{n\times n}(F)$, then a scalar λ is an eigenvalue of $A \Leftrightarrow \det(A - \lambda I_n) = 0$.

PROOF:

A scalar λ is an eigenvalue of A $\Leftrightarrow$ there exists $v \neq 0 \in F^n$ such that $Av = \lambda v$

$\Leftrightarrow$ $(A - \lambda I_n)v = 0$

$\Leftrightarrow$ $v \in N(A - \lambda I_n)$

$\Leftrightarrow$ $A - \lambda I_n$ is not invertible

$\Leftrightarrow$ $|A - \lambda I_n|$

Hence proved.

Note

- The Eigen values of a matrix are the zeros of its characteristic polynomial.
- If T is a linear operator on a finite-dimensional vector space V and β is an ordered basis for V, then λ is an eigenvalue of T if and only if λ is an eigenvalue of $[T]_\beta$.

Theorem

Let $A \in M_{n\times n}(F)$. Then

a. The characteristic polynomial of A is a polynomial of degree n with leading coefficient $(-1)^n$

b. The matrix A has at most n distinct eigenvalues.

Example 1

Find the eigenvalues of $A = \begin{pmatrix} 1 & 1 \\ 4 & 1 \end{pmatrix} \in M_{2\times 2}(R)$

Solution

The characteristic polynomial is $f(t) = \det(A - tI)$

$$\begin{aligned} f(t) &= \begin{vmatrix} 1-t & 1 \\ 4 & 1-t \end{vmatrix} \\ &= t^2 - 2t - 3 \\ &= (t-3)(t+1) \end{aligned}$$

$\therefore$ The eigenvalue of A are $3, -1$

Example 2

Let T be the linear operator on $P_2(R)$ defined by $T\big(f(x)\big) = f(x) + (x+1)f'(x)$ and β be the standard ordered basis for $P_2(R)$. Then compute characteristic polynomial and eigenvalue of T.

Solution

The standard basis for $P_2(R)$ is $\beta = \{1, x, x^2\}$. Then

$$\begin{aligned}
T(1) &= 1 + (1+1)(0) &&= 1 \\
T(x) &= x + (x+1)(1) &&= 1 + 2x \\
T(x^2) &= x^2 + (x+1)2x &&= 3x^2 + 2x
\end{aligned}$$

$$\therefore \qquad A = [T]_\beta = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}$$

The characteristic polynomial is

$$\begin{aligned} f(t) &= \det(A - tI) \\ &= \begin{vmatrix} 1-t & 1 & 0 \\ 0 & 2-t & 2 \\ 0 & 0 & 3-t \end{vmatrix} \\ &= (1-t)(2-t)(3-t) \\ f(t) &= -t^3 + 6t^2 - 11t + 6 \end{aligned}$$

$\therefore$ The eigenvalues of A are $1,\ 2$ and 3

Hence, The eigenvalues of T are $1,\ 2$ and 3

Theorem

Let T be a linear operator on a vector space V and let λ be an eigenvalue of T.

Then,

A vector $v \in V$ is an eigenvector of T corresponding to $\lambda \Leftrightarrow v \neq 0$ and $v \in N(T - \lambda I)$.

Example 1

Find the eigenvectors of the matrix $A = \begin{bmatrix} 1 & 1 \\ 4 & 1 \end{bmatrix}$ and check whether A is diagonalizable?

Solution

The characteristic polynomial is

$$\begin{aligned} f(t) &= |A - tI| \\ &= \begin{vmatrix} 1-t & 1 \\ 4 & 1-t \end{vmatrix} \\ &= (t+1)(t-3) \end{aligned}$$

$\therefore$ The eigenvalue of A are $\lambda_1 = -1, \lambda_2 = 3$

To find eigenvector

Case (1) when $\lambda_1 = -1$

Let $B_1 = A - \lambda_1 I = \begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$.

Then $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in R^2$ is an eigenvector corresponding to eigenvalue $\lambda_1 = -1 \Leftrightarrow X \neq 0$ and $X \in N(L_{B_1})$.

i.e., $X \neq 0$ and $\begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\Longrightarrow \quad 2x_1 + x_2 = 0$$

$$4x_1 + 2x_2 = 0$$

$$\Longrightarrow \quad x_2 = -2x_1$$

Let $x_1 = k,$ then $x_2 = -2k$

$$\therefore \qquad X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} k \\ -2k \end{bmatrix} = k \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Hence, $X = \left\{ k \begin{bmatrix} 1 \\ -2 \end{bmatrix} for\ some\ k \neq 0 \right\}.$

i.e., $\quad E_{\lambda_1} = \left\{ k \begin{bmatrix} 1 \\ -2 \end{bmatrix} for\ some\ k \neq 0 \right\}$

Case (2) when $\lambda_2 = 3$

Let $B_2 = A - \lambda_2 I = \begin{bmatrix} -2 & 1 \\ 4 & -2 \end{bmatrix}.$

Then,

$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in R^2$ is an eigenvector corresponding to eigenvalue $\lambda_2 = 3 \iff X \neq 0$ and $X \in N(L_{B_2})$.

i.e., $X \neq 0$ and $\begin{bmatrix} -2 & 1 \\ 4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\Longrightarrow$ $\quad -2x_1 + x_2 = 0$

$4x_1 - 2x_2 = 0$

$\Longrightarrow$ $\quad x_2 = 2x_1$

Let $x_1 = l$, then $x_2 = 2l$

$\therefore \quad X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} l \\ 2l \end{bmatrix} = l \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Hence, $X = \left\{ l \begin{bmatrix} 1 \\ 2 \end{bmatrix} for\ some\ l \neq 0 \right\}$

i.e., $\quad E_{\lambda_2} = \left\{ l \begin{bmatrix} 1 \\ 2 \end{bmatrix} for\ some\ l \neq 0 \right\}$

Let $\beta = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \end{bmatrix} \right\}$.

We have, β is Linearly independent and

$$span(\beta) = R^2.$$

$\therefore$ β is a basis for R^2

By the theorem "A linear operator T on a finite-dimensional vector space V is diagonalizable if and only if there exists an ordered basis β for V consisting of eigenvectors of T"

$\therefore$ L_A is diagonalizable.

Hence, A is diagonalizable.

Example 2

Let $T: R^3 \to R^3$ be a linear operator defined by

$T(x_1, x_2, x_3) = (2x_1 + x_2, x_2 - x_3, 2x_2 + 4x_3)$. Then find the all eigenvalues and a basis of each eigen space. Also check diagonalizability.

Solution

Let β be the standard basis for R^3.

Then, $[T]_\beta = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 2 & 4 \end{bmatrix}$

The characteristic polynomial is

$$f(\lambda) = |A - \lambda I|$$

$$= (\lambda - 2)^2(\lambda - 3)$$

$\therefore$ The eigenvalue of A are $2, 2\ and\ 3$

To find eigenspace

Let $B = [A - \lambda I]$. Then,

$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in R^3$ is an eigenvector corresponding to eigenvalue $\lambda \Leftrightarrow X \neq 0$ and $X \in N(L_B)$. i.e., $X \neq 0$

and $\begin{bmatrix} 2-\lambda & 1 & 0 \\ 0 & 1-\lambda & -1 \\ 0 & 2 & 4-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Case (1) when $\lambda_1 = 2$

$$\text{Let } B_1 = A - \lambda_1 I = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & -2 \end{bmatrix}.$$

Then, $X \neq 0$ and $\begin{bmatrix} 0 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\Longrightarrow \quad -x_2 = 0$$

$$x_2 + x_3 = 0$$

$$-2x_2 - 2x_3 = 0$$

Solving, we get $x_2 = 0, \; x_3 = 0$

Let $x_1 = a$. Then $X = \begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix}$

$$\therefore \qquad E_{\lambda_1} = \left\{ a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} for\ some\ a \neq 0 \right\}$$

Hence, the basis of the eigenspace E_{λ_1} is $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$.

Case (2) when $\lambda_2 = 3$

$$\text{Let } B_2 = A - \lambda_2 I = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & 1 \\ 0 & -2 & -1 \end{bmatrix}.$$

Then, $\quad X \neq 0$ and $\begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & 1 \\ 0 & -2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\Longrightarrow \qquad x_1 - x_2 = 0$$

$$2x_2 + x_3 = 0$$

$$-2x_2 - x_3 = 0$$

Solving, we get $\quad x_1 = 1,\ x_2 = 1,\ x_3 = -2$

i.e., $X = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$

$$\therefore \qquad E_{\lambda_2} = \left\{ \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \in R^3 \right\}$$

Hence, the basis of the eigenspace E_{λ_2} is $\begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$.

Since T has only two linearly independent eigenvectors, T is not diagonalizable.

Example 3

Let $T: R^3 \to R^3$ be a linear operator defined by

$T(x_1, x_2, x_3) = (2x_1 + x_2, x_2 - x_3, 2x_2 + 4x_3)$. Then find the all eigenvalues and a basis of each eigen space. Also check diagonalizability?

Solution

Let β be the standard basis for R^3.

Then, $A = [T]_\beta = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 2 & 4 \end{bmatrix}$

The characteristic polynomial is

$$f(\lambda) = |A - \lambda I|$$

$$= (\lambda - 2)^2(\lambda - 3)$$

$\therefore$ The eigenvalue of A are $2, 2\ and\ 3$

To find eigenspace

Let $B = [A - \lambda I]$. Then,

$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in R^3$ is an eigenvector corresponding to eigenvalue $\lambda \Leftrightarrow X \neq 0$ and $X \in N(L_B)$. i.e., $X \neq 0$

and $\begin{bmatrix} 2-\lambda & 1 & 0 \\ 0 & 1-\lambda & -1 \\ 0 & 2 & 4-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Case (1) when λ_1 =2

$$\text{Let } B_1 = A - \lambda_1 I = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & -2 \end{bmatrix}.$$

Then, $X \neq 0$ and $\begin{bmatrix} 0 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\Longrightarrow \quad -x_2 = 0$$

$$x_2 + x_3 = 0$$

$$-2x_2 - 2x_3 = 0$$

Solving, we get $x_2 = 0,\ x_3 = 0$

Let $x_1 = a$. Then $X = \begin{bmatrix} a \\ 0 \\ 0 \end{bmatrix}$

$$\therefore \qquad E_{\lambda_1} = \left\{ a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} for\ some\ a \neq 0 \right\}$$

Hence, the basis of the eigenspace E_{λ_1} is $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$.

Case (2) when $\lambda_2 = 3$

$$\text{Let } B_2 = A - \lambda_2 I = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & 1 \\ 0 & -2 & -1 \end{bmatrix}.$$

Then, $\quad X \neq 0$ and $\begin{bmatrix} 1 & -1 & 0 \\ 0 & 2 & 1 \\ 0 & -2 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\Longrightarrow \qquad x_1 - x_2 = 0$$

$$2x_2 + x_3 = 0$$

$$-2x_2 - x_3 = 0$$

Solving, we get $\quad x_1 = 1,\ x_2 = 1,\ x_3 = -2$

i.e., $X = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$

$$\therefore \qquad E_{\lambda_2} = \left\{ \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \in R^3 \right\}$$

Hence, the basis of the eigenspace E_{λ_2} is $\begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}$.

Since T has only two linearly independent eigenvectors, T is not diagonalizable.

Definition

A polynomial $f(t)$ in $P(F)$ **splits over F** if there are scalars $c, a_1, a_2, \dots, a_n$ (not necessary distinct) in F such that $f(t) = c(t - a_1)(t - a_2)\cdots(t - a_n)$

Example

1. $f(t) = t^2 - 1 = (t + 1)(t - 1)$ splits over R
2. $f(t) = t^2 + 1 = (t + i)(t - i)$ does not split over R, but $f(t)$ splits over C

Definition

Let λ be an eigenvalue of a linear operator or matrix with characteristic polynomial $f(t)$. Then, the **algebraic or multiplicity of $\boldsymbol{\lambda}$** is the largest positive integer k for which $(t-\lambda)^k$ is a factor of $f(t)$.

Example

Let $A = \begin{pmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{pmatrix} \in \mathrm{M}_{3\times 3}(\mathrm{R})$. Then find the multiplicity of each eigenvalues.

Solution :

$$f(t) = \begin{vmatrix} 3-\mathrm{t} & 1 & 0 \\ 0 & 3-\mathrm{t} & 0 \\ 0 & 0 & 4-\mathrm{t} \end{vmatrix}$$

$$= -(\mathrm{t}-3)^2(\mathrm{t}-4)$$

$\therefore f(t)$ is split over R.

Eigen values of A are 3 & 4

The multiplicity of $\lambda_1 = 3$ is 2

The multiplicity of $\lambda_2 = 4$ is 1

Test for Diagonalization

Let T be a linear operator on an n-dimensional vector space V. Then T is diagonalizable if and only if both of the following conditions hold.

1. The characteristic polynomial of T splits.

2. For each eigenvalue λ of T, the multiplicity of λ equals $n - rank(T - \lambda I)$.

Example 1

Let T be a linear operator on $P_2(R)$ defined by

$$T\big(f(x)\big) = f(1) + f'(0)x + \big(f'(0) + f''(0)\big)x^2.$$

Test the diagonalizability of T**.** If so, find the diagonal matrix.

Solution

Let β be the standard basis for$P_2(R)$.

Then, $A = [T]_\beta = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix}$

The characteristic polynomial is

$$f(t) \; = \; |A - tI|$$

$$= -\,(t-1)^2(t-2)$$

$\therefore \;\; f(t)$ is split over R

Also, The eigenvalue of A are $1, 1 \; and \; 2$

The multiplicity of $\lambda_1 = 1$ is 2

The multiplicity of $\lambda_2 = 2$ is 1

For $\boldsymbol{\lambda_1 = 1}$, we have $A - \lambda_1 I = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$

$$rank(A - \lambda_1 I) = 1$$

$$n - rank(A - \lambda_1 I) = 3 - 1 = 2$$

$\therefore$ The multiplicity of $\lambda_1 = n - rank(A - \lambda_1 I)$

For $\boldsymbol{\lambda_2 = 2}$, we have $A - \lambda_2 I = \begin{bmatrix} -1 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$

$$rank(A - \lambda_2 I) = 2$$

$$n - rank(A - \lambda_2 I) = 3 - 2 = 1$$

$\therefore$ The multiplicity of $\lambda_2 = n - rank(A - \lambda_2 I)$

Hence, T is diagonalizable

To find diagonal matrix:

Let $B = [A - \lambda I]$. Then,

$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$ is an eigenvector corresponding to eigenvalue $\lambda \Leftrightarrow X \neq 0$ and $X \in N(L_B)$.

i.e., $X \neq 0$ and $\begin{bmatrix} 1-\lambda & 1 & 1 \\ 0 & 1-\lambda & 0 \\ 0 & 1 & 2-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Case (1) when $\lambda_1 = 1$

Let $B_1 = A - \lambda_1 I = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix}$.

Then, $X \neq 0$ and $\begin{bmatrix} 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\Longrightarrow \quad x_2 + x_3 = 0$$

Solving, we get $x_3 = -x_2$

Let $x_1 = a, x_2 = b,$. Then $X = \begin{bmatrix} a \\ b \\ -b \end{bmatrix}$

$$\therefore \qquad E_{\lambda_1} = \left\{ a \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} for\ some\ a, b \neq 0 \right\}$$

Hence, the basis of the eigenspace E_{λ_1} are $\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$.

Case (2) when $\lambda_2 = 2$

$$\text{Let } B_2 = A - \lambda_2 I = \begin{bmatrix} -1 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Then, $\quad X \neq 0$ and $\begin{bmatrix} -1 & 1 & 1 \\ 0 & -1 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\Longrightarrow \qquad -x_1 + x_2 + x_3 = 0$$

$$-x_2 = 0$$

$$x_2 = 0$$

Solving, we get $\quad x_2 = 0,\ x_3 = x_1$

Let $x_1 = a$. Then $X = \begin{bmatrix} a \\ 0 \\ a \end{bmatrix}$

$$\therefore \qquad E_{\lambda_2} = \left\{ a \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} for\ some\ a \neq 0 \right\}$$

Hence, the basis of the eigenspace E_{λ_2} is $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$.

Let $B = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right\}$.

Then B is an ordered basis for R^3 consisting of eigenvectors of A.

The vectors in B are the coordinate vectors relative to β of the vectors in the set $\gamma = \{1, x - x^2, 1 +$

By similarity transformation,

Let $Q = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$.

Then, $D \quad = \quad Q^{-1}AQ$

$$= \begin{bmatrix} 1 & -1 & -1 \\ 0 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$\therefore$ The diagonal matrix is $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

Example 2

Test diagonalizability of the matrix $A = \begin{bmatrix} 3 & 1 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix} \in M_{3\times3}(R)$

Solution

The characteristic polynomial is

$$f(t) = |A - tI|$$

$$= -(t-3)^2(t-4)$$

$\therefore\ f(t)$ is split over R

Also, The eigenvalue of A are 3 *and* 4

The multiplicity of $\lambda_1 = 3$ is 2

The multiplicity of $\lambda_2 = 4$ is 1

For $\lambda_1 = 3$, we have $A - \lambda_1 I = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$rank(A - \lambda_1 I) = 2$$

$$n - rank(A - \lambda_1 I) = 3 - 2 = 1$$

$\therefore$ The multiplicity of $\lambda_1 \neq n - rank(A - \lambda_1 I)$

For $\lambda_2 = 4$, we have $A - \lambda_2 I = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

$$rank(A - \lambda_2 I) = 2$$

$$n - rank(A - \lambda_2 I) = 3 - 2 = 1$$

$\therefore$ The multiplicity of $\lambda_2 = n - rank(A - \lambda_2 I)$

Hence T not is diagonalizable.

APPLICATION OF DIAGONALIZATIONS IN LINEAR DIFFERENTIAL EQUATIONS

Consider the system of equations

$$x_1' = a_{11}x_1 + a_{12}x_2 + a_{13}x_3$$

$$x_2' = a_{21}x_1 + a_{22}x_2 + a_{23}x_3$$

$$x_3' = a_{31}x_1 + a_{32}x_2 + a_{33}x_3$$

where, $x_i = x_i(t)\ \forall\ i$, is a differentiable real-valued function of the real variable t.

Clearly, this system has a solution (trivial solution, i.e., each $x_i(t)$ is the zero function).

We determine all of the solutions to this system.

Let $x: R \longrightarrow R^3$ be the function defined by $x(t) = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}$

The derivative of x, denoted $x'(t)$ and is defined by $x'(t) = \begin{bmatrix} x_1'(t) \\ x_2'(t) \\ x_3'(t) \end{bmatrix}$

Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ be the coefficient matrix of the given system, then the system as the matrix equation $x' = Ax$ -----(1)

If A is diagonalizable, then we have $Q^{-1}AQ = D$.

$\Longrightarrow A = QDQ^{-1}$

Substitue in (1), we get

$$x' = QDQ^{-1}x$$

$$\Longrightarrow \quad Q^{-1}x' = DQ^{-1}x$$

The function $y: R \longrightarrow R^3$ defined by $y(t) = Q^{-1}x(t)$

The derivative of y is defined by $y' = Q^{-1}x'$

$$= DQ^{-1}x$$

$$y' = Dy \qquad \text{-----(2)}$$

Since D is a diagonal matrix, the system $y' = Dy$ is easy to solve.

Let $y(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{bmatrix}$

From (2),

$$y' = Dy$$

$$\begin{bmatrix} y_1'(t) \\ y_2'(t) \\ y_3'(t) \end{bmatrix} = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix} \begin{bmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{bmatrix}$$

$$= \begin{bmatrix} \lambda_1 y_1(t) \\ \lambda_2 y_2(t) \\ \lambda_3 y_3(t) \end{bmatrix}$$

$$\Longrightarrow \quad y_1'(t) = \lambda_1 y_1(t)$$

$$y_2'(t) = \lambda_2 y_2(t)$$

$$y_3'(t) = \lambda_3 y_3(t)$$

These equations are independent of each other and thus can be solved individually.

The general solution to these equations is

$$y_1(t) = c_1 e^{\lambda_1 t}$$

$$y_2(t) = c_2 e^{\lambda_2 t}$$

$$y_3(t) = c_3 e^{\lambda_3 t}$$

where c_1, c_2, c_3 are arbitrary constants.

From $y(t) = Q^{-1}x(t)$, we have $x(t) = Qy(t)$,

which is the general solution of the given system.

Example

Find the general solution to each system of differential equations

$$x_1' = 3x_1 + x_2 + x_3$$

$$x_2' = 2x_1 + 4x_2 + 2x_3$$

$$x_3' = -x_1 - x_2 + x_3$$

Solution

Let the given system as $x' = Ax$

where, $x' = \begin{bmatrix} x_1'(t) \\ x_2'(t) \\ x_3'(t) \end{bmatrix}$, $A = \begin{bmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ -1 & -1 & 1 \end{bmatrix}$, $x = \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix}$

The characteristic polynomial is $f(t) = \det(A - \lambda I)$

$$= \begin{vmatrix} 3-t & 1 & 1 \\ 2 & 4-t & 2 \\ -1 & -1 & 1-t \end{vmatrix}$$

$$= (t-2)(t-2)(t-4)$$

To find eigen values:

$$f(t) = 0 \quad \Longrightarrow \quad (t-2)(t-2)(t-4) = 0$$

$$\Longrightarrow \quad t = 2, 2, 4$$

$\therefore$ The eigen values are $2, 2, 4$

To find eigenvector

Let $B = [A - \lambda I]$. Then, $Z = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} \in R^3$ is an eigenvector corresponding to eigenvalue $\lambda \Longleftrightarrow Z \neq 0$

and $Z \in N(L_B)$.

i.e., $\quad Z \neq 0$ and $\begin{bmatrix} 3-\lambda & 1 & 1 \\ 2 & 4-\lambda & 2 \\ -1 & -1 & 1-\lambda \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

Case (1) when $\lambda_1 = 2$

Let $B_1 = A - \lambda_1 I = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix}$.

Then, $\quad Z \neq 0$ and $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\Rightarrow \quad z_1 + z_2 + z_3 = 0;\ 2z_1 + 2z_2 + 2z_3 = 0;\ -z_1 - z_2 - z_3 = 0$$

$$\Rightarrow \quad z_1 + z_2 + z_3 = 0$$

Put $z_1 = 0$. Then $z_2 = -z_3$

Let $z_3 = 1$, then $z_2 = -1$

$$\therefore\ Z_1 = \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

Put $z_2 = 0$. Then $z_1 = -z_3$

Let $z_3 = 1$, then $z_1 = -1$

$$\therefore Z_2 = \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

Case (2) when $\lambda_2 = 4$

Let $B_2 = A - \lambda_2 I = \begin{bmatrix} -1 & 1 & 1 \\ 2 & 0 & 2 \\ -1 & -1 & -3 \end{bmatrix}$.

Then, $Z \neq 0$ and $\begin{bmatrix} -1 & 1 & 1 \\ 2 & 0 & 2 \\ -1 & -1 & -3 \end{bmatrix} \begin{bmatrix} z_1 \\ z_2 \\ z_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$\Longrightarrow \quad -z_1 + z_2 + z_3 = 0$$

$$2z_1 + 2z_3 = 0$$

$$-z_1 - z_2 - 3z_3 = 0$$

$$\therefore \; Z_3 = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

Let $B = \left\{ \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \right\}$.

Then, B is an ordered basis for R^3 consisting of eigenvectors of A.

$\therefore \quad A$ is diagonalizable

Let $Q = \begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & 1 & -1 \end{bmatrix}$.

Then $\quad D = Q^{-1}AQ$

$$= \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Now, put $A = QDQ^{-1}$ in x', we get

$x'(t) = (QDQ^{-1})x \Longrightarrow Q^{-1}x'(t) = DQ^{-1}x$ --------(*)

Define $y: R \to R^3$ by $y = Q^{-1}x(t)$, then by equation (*) $y' = Dy$

$$\Longrightarrow \quad \begin{bmatrix} y_1'(t) \\ y_2'(t) \\ y_3'(t) \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{bmatrix} \begin{bmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{bmatrix}$$

$$= \begin{bmatrix} 2y_1(t) \\ 2y_2(t) \\ 4y_3(t) \end{bmatrix}$$

$\therefore \quad y_1'(t) = 2y_1(t)$

$$y_2'(t) = 2y_2(t)$$

$$y_3'(t) = 4y_3(t)$$

Solving these equations, we get

$$y_1(t) = c_1e^{2t}$$

$$y_2(t) = c_2e^{2t}$$

$$y_3(t) = c_3e^{4t}$$

where c_1, c_2 *and* c_3 are arbitrary constants.

Now, $x(t) = Qy(t)$

$$\Rightarrow \quad \begin{bmatrix} x_1(t) \\ x_2(t) \\ x_3(t) \end{bmatrix} = \begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} y_1(t) \\ y_2(t) \\ y_3(t) \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 2 \\ 1 & 1 & -1 \end{bmatrix} \begin{bmatrix} c_1 e^{2t} \\ c_2 e^{2t} \\ c_3 e^{4t} \end{bmatrix}$$

$$= \begin{bmatrix} -c_1 e^{2t} + c_3 e^{4t} \\ -c_2 e^{2t} + 2c_3 e^{4t} \\ c_1 e^{2t} + c_2 e^{2t} - c_3 e^{4t} \end{bmatrix}$$

$$= \begin{bmatrix} -c_1 e^{2t} \\ 0 \\ c_1 e^{2t} \end{bmatrix} + \begin{bmatrix} 0 \\ -c_2 e^{2t} \\ c_2 e^{2t} \end{bmatrix} + \begin{bmatrix} c_3 e^{4t} \\ 2c_3 e^{4t} \\ -c_3 e^{4t} \end{bmatrix}$$

$$x(t) = \begin{bmatrix} -c_1 e^{2t} \\ 0 \\ c_1 e^{2t} \end{bmatrix} + \begin{bmatrix} 0 \\ -c_2 e^{2t} \\ c_2 e^{2t} \end{bmatrix} + \begin{bmatrix} c_3 e^{4t} \\ 2c_3 e^{4t} \\ -c_3 e^{4t} \end{bmatrix}$$

$$= c_1 e^{2t} \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + c_2 e^{2t} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} + c_3 e^{4t} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$= e^{2t} \left\{ c_1 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right\} + c_3 e^{4t} \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$\therefore$ The general solution is

$$x(t) = e^{2t} z_1 + e^{4t} z_2, where\ z_1 \in E\lambda_1\ and\ z_2 \in E\lambda_2$$

PROBLEM:

Find the general solution by the system of differential equations.

$$x_1'(t) = x_1(t) + x_2(t)$$

$$x_2'(t) = 3x_1(t) - x_2(t)$$